

```
In [1]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
In [2]: %cd /content/drive/MyDrive/content/drive/MyDrive/IDS561
```

/content/drive/MyDrive/content/drive/MyDrive/IDS561

```
In [3]: #Complete the installation process
!apt-get install openjdk-8-jdk-headless -qq > /dev/null
```

```
In [4]: !ls
```

```
!tar -xvf spark-3.5.1-bin-hadoop3.tgz
!ls
```

```
Assignment3          final_experiment.csv      spark-3.5.1-bin-hadoop3.tgz
daily_active_users.csv iris.data                  spark-3.5.1-bin-hadoop3.tgz.3
experiment.txt        spark-3.4.2-bin-hadoop3
experiment_user.csv   spark-3.5.1-bin-hadoop3
spark-3.5.1-bin-hadoop3/
spark-3.5.1-bin-hadoop3/sbin/
spark-3.5.1-bin-hadoop3/sbin/spark-config.sh
spark-3.5.1-bin-hadoop3/sbin/stop-slave.sh
spark-3.5.1-bin-hadoop3/sbin/stop-mesos-dispatcher.sh
spark-3.5.1-bin-hadoop3/sbin/start-workers.sh
spark-3.5.1-bin-hadoop3/sbin/start-slaves.sh
spark-3.5.1-bin-hadoop3/sbin/start-all.sh
spark-3.5.1-bin-hadoop3/sbin/stop-all.sh
spark-3.5.1-bin-hadoop3/sbin/workers.sh
spark-3.5.1-bin-hadoop3/sbin/start-mesos-dispatcher.sh
spark-3.5.1-bin-hadoop3/sbin/spark-daemon.sh
spark-3.5.1-bin-hadoop3/sbin/decommission-worker.sh
spark-3.5.1-bin-hadoop3/sbin/slaves.sh
spark-3.5.1-bin-hadoop3/sbin/stop-mesos-shuffle-service.sh
spark-3.5.1-bin-hadoop3/sbin/stop-history-server.sh
spark-3.5.1-bin-hadoop3/sbin/stop-worker.sh
spark-3.5.1-bin-hadoop3/sbin/decommission-slave.sh
spark-3.5.1-bin-hadoop3/sbin/stop-thriftserver.sh
spark-3.5.1-bin-hadoop3/sbin/start-worker.sh
spark-3.5.1-bin-hadoop3/sbin/stop-slaves.sh
spark-3.5.1-bin-hadoop3/sbin/start-connect-server.sh
spark-3.5.1-bin-hadoop3/sbin/start-thriftserver.sh
spark-3.5.1-bin-hadoop3/sbin/start-history-server.sh
spark-3.5.1-bin-hadoop3/sbin/stop-connect-server.sh
spark-3.5.1-bin-hadoop3/sbin/spark-daemons.sh
spark-3.5.1-bin-hadoop3/sbin/start-slave.sh
spark-3.5.1-bin-hadoop3/sbin/start-master.sh
spark-3.5.1-bin-hadoop3/sbin/start-mesos-shuffle-service.sh
spark-3.5.1-bin-hadoop3/sbin/stop-master.sh
spark-3.5.1-bin-hadoop3/sbin/stop-workers.sh
spark-3.5.1-bin-hadoop3/licenses/
spark-3.5.1-bin-hadoop3/licenses/LICENSE-automaton.txt
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spark-3.5.1-bin-hadoop3/jars/hive-shims-0.23-2.3.9.jar
spark-3.5.1-bin-hadoop3/jars/kubernetes-model-autoscaling-6.7.2.jar
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spark-3.5.1-bin-hadoop3/jars/netty-resolver-4.1.96.Final.jar

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r.py
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spark-3.5.1-bin-hadoop3/python/pyspark/sql/tests/connect/streaming/
spark-3.5.1-bin-hadoop3/python/pyspark/sql/tests/connect/streaming/test_parity_streamin
g.py
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atch.py
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py
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with_state.py
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agg.py
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py
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p.py
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spark-3.5.1-bin-hadoop3/python/pyspark/sql/tests/connect/test_parity_pandas_map.py
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spark-3.5.1-bin-hadoop3/python/pyspark/sql/tests/streaming/test_streaming_foreach_batch.

py
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spark-3.5.1-bin-hadoop3/python/pyspark/sql/pandas/_typing/protocols/series.pyi
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ply.py
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data.py
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index.py
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series.py
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plotly.py
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_plotly.py
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matplotlib.py
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spark-3.5.1-bin-hadoop3/python/pyspark/pandas/tests/connect/indexes/test_parity_reindex.
py
spark-3.5.1-bin-hadoop3/python/pyspark/pandas/tests/connect/indexes/test_parity_reset_in

dex.py
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spark-3.5.1-bin-hadoop3/python/pyspark/pandas/tests/connect/indexes/test_parity_indexin
g.py
spark-3.5.1-bin-hadoop3/python/pyspark/pandas/tests/connect/indexes/test_parity_datetim
e.py
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y
spark-3.5.1-bin-hadoop3/python/pyspark/pandas/tests/connect/test_parity_series_string.py
spark-3.5.1-bin-hadoop3/python/pyspark/pandas/tests/connect/test_parity_generic_function
s.py
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es_groupby_rolling.py
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n.py
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n.py
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y
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e.py
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y
spark-3.5.1-bin-hadoop3/python/pyspark/pandas/tests/connect/series/test_parity_arg_ops.p
y
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ata.py
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py
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spark-3.5.1-bin-hadoop3/python/pyspark/pandas/tests/data_type_ops/test_datetime_ops.py

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spark-3.5.1-bin-hadoop3/python/pyspark/mllib/tests/test_algorithms.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/tests/test_feature.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/tests/test_util.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/tests/test_linalg.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/tests/test_stat.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/tests/__init__.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/linalg/
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/linalg/distributed.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/linalg/__init__.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/evaluation.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/stat/
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/stat/distribution.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/stat/KernelDensity.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/stat/test.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/stat/__init__.py

spark-3.5.1-bin-hadoop3/python/pyspark/mllib/stat/_statistics.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/classification.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/feature.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/fpm.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/__init__.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/tree.py
spark-3.5.1-bin-hadoop3/python/pyspark/mllib/random.py
spark-3.5.1-bin-hadoop3/python/pyspark/broadcast.py
spark-3.5.1-bin-hadoop3/python/pyspark/daemon.py
spark-3.5.1-bin-hadoop3/python/pyspark/traceback_utils.py
spark-3.5.1-bin-hadoop3/python/pyspark/serializers.py
spark-3.5.1-bin-hadoop3/python/pyspark/errors/
spark-3.5.1-bin-hadoop3/python/pyspark/errors/tests/
spark-3.5.1-bin-hadoop3/python/pyspark/errors/tests/test_errors.py
spark-3.5.1-bin-hadoop3/python/pyspark/errors/tests/__init__.py
spark-3.5.1-bin-hadoop3/python/pyspark/errors/error_classes.py
spark-3.5.1-bin-hadoop3/python/pyspark/errors/utils.py
spark-3.5.1-bin-hadoop3/python/pyspark/errors/exceptions/
spark-3.5.1-bin-hadoop3/python/pyspark/errors/exceptions/connect.py
spark-3.5.1-bin-hadoop3/python/pyspark/errors/exceptions/captured.py
spark-3.5.1-bin-hadoop3/python/pyspark/errors/exceptions/base.py
spark-3.5.1-bin-hadoop3/python/pyspark/errors/exceptions/__init__.py
spark-3.5.1-bin-hadoop3/python/pyspark/errors/__init__.py
spark-3.5.1-bin-hadoop3/python/pyspark/shell.py
spark-3.5.1-bin-hadoop3/python/pyspark/python/
spark-3.5.1-bin-hadoop3/python/pyspark/python/pyspark/
spark-3.5.1-bin-hadoop3/python/pyspark/python/pyspark/shell.py
spark-3.5.1-bin-hadoop3/python/pyspark/taskcontext.py
spark-3.5.1-bin-hadoop3/python/pyspark/files.py
spark-3.5.1-bin-hadoop3/python/pyspark/__init__.py
spark-3.5.1-bin-hadoop3/python/pyspark/install.py
spark-3.5.1-bin-hadoop3/python/pyspark/_globals.py
spark-3.5.1-bin-hadoop3/python/pyspark/shuffle.py
spark-3.5.1-bin-hadoop3/python/pyspark/java_gateway.py
spark-3.5.1-bin-hadoop3/python/pyspark/find_spark_home.py
spark-3.5.1-bin-hadoop3/RELEASE
spark-3.5.1-bin-hadoop3/R/
spark-3.5.1-bin-hadoop3/R/lib/
spark-3.5.1-bin-hadoop3/R/lib/SparkR/
spark-3.5.1-bin-hadoop3/R/lib/SparkR/Meta/
spark-3.5.1-bin-hadoop3/R/lib/SparkR/Meta/nsInfo.rds
spark-3.5.1-bin-hadoop3/R/lib/SparkR/Meta/hsearch.rds
spark-3.5.1-bin-hadoop3/R/lib/SparkR/Meta/vignette.rds
spark-3.5.1-bin-hadoop3/R/lib/SparkR/Meta/Rd.rds
spark-3.5.1-bin-hadoop3/R/lib/SparkR/Meta/package.rds
spark-3.5.1-bin-hadoop3/R/lib/SparkR/Meta/links.rds
spark-3.5.1-bin-hadoop3/R/lib/SparkR/Meta/features.rds
spark-3.5.1-bin-hadoop3/R/lib/SparkR/doc/
spark-3.5.1-bin-hadoop3/R/lib/SparkR/doc/index.html
spark-3.5.1-bin-hadoop3/R/lib/SparkR/doc/sparkr-vignettes.html
spark-3.5.1-bin-hadoop3/R/lib/SparkR/doc/sparkr-vignettes.Rmd
spark-3.5.1-bin-hadoop3/R/lib/SparkR/doc/sparkr-vignettes.R
spark-3.5.1-bin-hadoop3/R/lib/SparkR/help/
spark-3.5.1-bin-hadoop3/R/lib/SparkR/help/SparkR.rdx
spark-3.5.1-bin-hadoop3/R/lib/SparkR/help/paths.rds
spark-3.5.1-bin-hadoop3/R/lib/SparkR/help/SparkR.rdb
spark-3.5.1-bin-hadoop3/R/lib/SparkR/help/AnIndex
spark-3.5.1-bin-hadoop3/R/lib/SparkR/help/aliases.rds
spark-3.5.1-bin-hadoop3/R/lib/SparkR/NAMESPACE
spark-3.5.1-bin-hadoop3/R/lib/SparkR/tests/
spark-3.5.1-bin-hadoop3/R/lib/SparkR/tests/testthat/
spark-3.5.1-bin-hadoop3/R/lib/SparkR/tests/testthat/test_basic.R
spark-3.5.1-bin-hadoop3/R/lib/SparkR/INDEX
spark-3.5.1-bin-hadoop3/R/lib/SparkR/DESCRIPTION
spark-3.5.1-bin-hadoop3/R/lib/SparkR/profile/
spark-3.5.1-bin-hadoop3/R/lib/SparkR/profile/general.R

```

spark-3.5.1-bin-hadoop3/R/lib/SparkR/profile/shell.R
spark-3.5.1-bin-hadoop3/R/lib/SparkR/worker/
spark-3.5.1-bin-hadoop3/R/lib/SparkR/worker/daemon.R
spark-3.5.1-bin-hadoop3/R/lib/SparkR/worker/worker.R
spark-3.5.1-bin-hadoop3/R/lib/SparkR/html/
spark-3.5.1-bin-hadoop3/R/lib/SparkR/html/00Index.html
spark-3.5.1-bin-hadoop3/R/lib/SparkR/html/R.css
spark-3.5.1-bin-hadoop3/R/lib/SparkR/R/
spark-3.5.1-bin-hadoop3/R/lib/SparkR/R/SparkR.rdx
spark-3.5.1-bin-hadoop3/R/lib/SparkR/R/SparkR.rdb
spark-3.5.1-bin-hadoop3/R/lib/SparkR/R/SparkR
spark-3.5.1-bin-hadoop3/R/lib/sparkr.zip
Assignment3          final_experiment.csv      spark-3.5.1-bin-hadoop3.tgz
daily_active_users.csv iris.data          spark-3.5.1-bin-hadoop3.tgz.3
experiment.txt        spark-3.4.2-bin-hadoop3
experiment_user.csv   spark-3.5.1-bin-hadoop3

```

```
In [5]: !pip install -q findspark
```

```
In [6]: import os
os.environ["JAVA_HOME"] = "/usr/lib/jvm/java-8-openjdk-amd64"
os.environ["SPARK_HOME"] = "/content/drive/MyDrive/content/drive/MyDrive/IDS561/spark-3.
```

```
In [7]: import findspark
findspark.init()
from pyspark.sql import SparkSession
spark = SparkSession.builder.master("local[*]").getOrCreate()
```

```
In [8]: df = spark.read.options(delimiter=',', inferSchema=True, header=True).csv('/content/driv
```

```
In [9]: df.limit(3).toPandas().style.hide(axis="index")
```

```
Out[9]:
```

Store	Dept	Date	Weekly_Sales	IsHoliday
1	1	2010-02-05	24924.500000	False
1	1	2010-02-12	46039.490000	True
1	1	2010-02-19	41595.550000	False

```
In [11]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
In [12]: import pyspark.sql.functions as f
import pyspark
```

```
In [13]: mn_dt = df.select('Date').agg(f.min('Date')).collect()[0][0]
mx_dt = df.select('Date').agg(f.max('Date')).collect()[0][0]
print('Total Rows: ', df.count())
print('Min Date: ', mn_dt)
print('Max Date: ', mx_dt)
```

```
Total Rows: 421570
Min Date: 2010-02-05
Max Date: 2012-10-26
```

```
In [15]: stores = spark.read.options(delimiter=',', inferSchema=True, header=True).csv('/content/
stores.limit(3).toPandas().style.hide(axis="index")
```

```
Out[15]:
```

Store	Type	Size
-------	------	------

```
1      A   151315
2      A   202307
3      B   37392
```

```
In [16]: print('Total Rows: ', stores.count())
print('Total Stores: ', stores.select('Store').distinct().count())
stores.select('Type').distinct().show()
```

```
Total Rows:  45
Total Stores: 45
+----+
|Type|
+----+
|  B |
|  C |
|  A |
+----+
```

```
In [18]: features = spark.read.options(delimiter=',', inferSchema=True, header=True).csv('/content/
features.limit(3).toPandas().style.hide(axis="index")
```

```
Out[18]:
```

	Store	Date	Temperature	Fuel_Price	MarkDown1	MarkDown2	MarkDown3	MarkDown4	MarkDown5	
	1	2010-02-05	42.310000	2.572000	NA	NA	NA	NA	NA	211.096
	1	2010-02-12	38.510000	2.548000	NA	NA	NA	NA	NA	211.242
	1	2010-02-19	39.930000	2.514000	NA	NA	NA	NA	NA	211.289

```
In [19]: mn_dt = features.select('Date').agg(f.min('Date')).collect()[0][0]
mx_dt = features.select('Date').agg(f.max('Date')).collect()[0][0]
print('Total Rows: ', features.count())
print('Min Date: ', mn_dt)
print('Max Date: ', mx_dt)
```

```
Total Rows:  8190
Min Date:  2010-02-05
Max Date:  2013-07-26
```

```
In [20]: # Converting 'NA' strings in MarkDown columns to null values
for i in range(1,6):
    features = features.withColumn(
        f'MarkDown{i}', f.when(f.col(f'MarkDown{i}')=='NA', None).otherwise(
            f.col(f'MarkDown{i}')).cast('float'))
```

```
In [21]: pdf = df.join(
    stores, on='Store', how='inner').join(
    features.drop('IsHoliday'), on=['Date', 'Store'], how='left').toPandas()
pdf['Date'] = pd.to_datetime(pdf['Date'])
pdf.set_index('Date', inplace=True)
pdf["CPI"] = pd.to_numeric(pdf["CPI"], downcast="float")
pdf["Unemployment"] = pd.to_numeric(pdf["Unemployment"], downcast="float")
pdf.head()
```

```
Out[21]:
```

	Store	Dept	Weekly_Sales	IsHoliday	Type	Size	Temperature	Fuel_Price	MarkDown1	MarkDown2
2010-02-05	1	1	24924.50	False	A	151315	42.31	2.572	NaN	NaN
2010-	1	1	46039.49	True	A	151315	38.51	2.548	NaN	NaN

02-12

2010-02-19	1	1	41595.55	False	A	151315	39.93	2.514	NaN	NaN
2010-02-26	1	1	19403.54	False	A	151315	46.63	2.561	NaN	NaN
2010-03-05	1	1	21827.90	False	A	151315	46.50	2.625	NaN	NaN

In [22]: pdf.dtypes

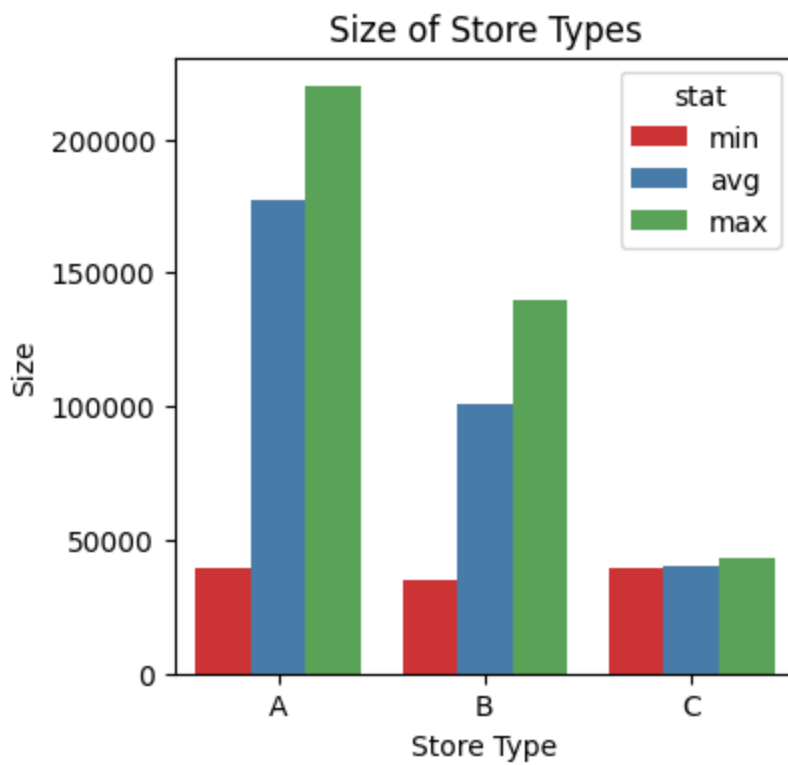
Out[22]:

Store	int32
Dept	int32
Weekly_Sales	float64
IsHoliday	bool
Type	object
Size	int32
Temperature	float64
Fuel_Price	float64
Markdown1	float32
Markdown2	float32
Markdown3	float32
Markdown4	float32
Markdown5	float32
CPI	float32
Unemployment	float32
dtype:	object

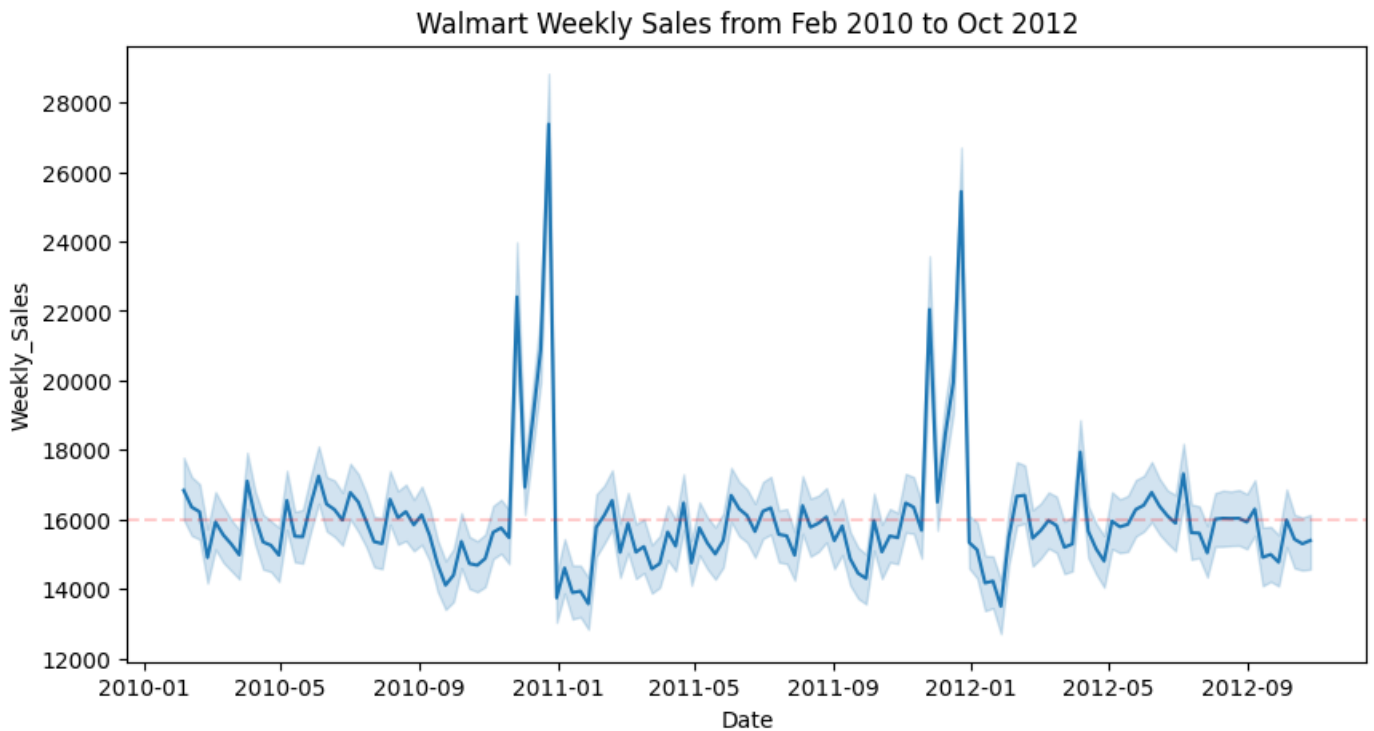
In [23]:

```
# Checking min, max, average of store by Type
stdf = stores.groupBy('Type').agg(
    f.min('Size').alias('min'),
    f.round(f.avg('Size'),2).alias('avg'),
    f.max('Size').alias('max')).orderBy('Type').toPandas()

m_df = pd.melt(stdf, id_vars="Type", var_name="stat", value_name="Size")
fig, ax = plt.subplots(figsize=(4,4))
sns.barplot(x='Type', y='Size', hue='stat', data=m_df, ax=ax, palette='Set1').set_title(
plt.xlabel('Store Type')
plt.show()
```

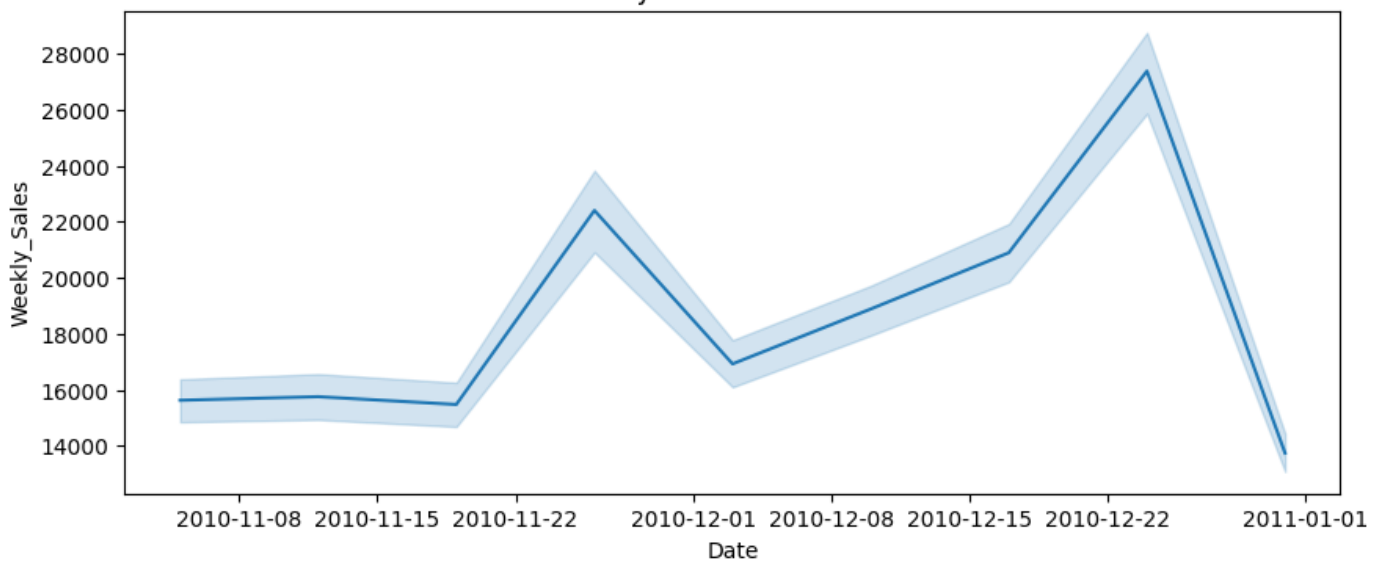


```
In [24]: plt.figure(figsize=(10,5))
sns.lineplot(data=pdf,
              x='Date',
              y='Weekly_Sales',
              ).set_title('Walmart Weekly Sales from Feb 2010 to Oct 2012')
plt.axhline(pdf['Weekly_Sales'].mean(), color='r', alpha=0.2, linestyle='--')
plt.show()
```



```
In [25]: plt.figure(figsize=(10,4))
sns.lineplot(
    data=pdf.loc[(pdf.index > '2010-11-01') & (pdf.index < '2011-01-01')],
    x='Date', y='Weekly_Sales'
).set_title('Walmart Weekly Sales from Nov to Dec 2010')
plt.show()
```

Walmart Weekly Sales from Nov to Dec 2010



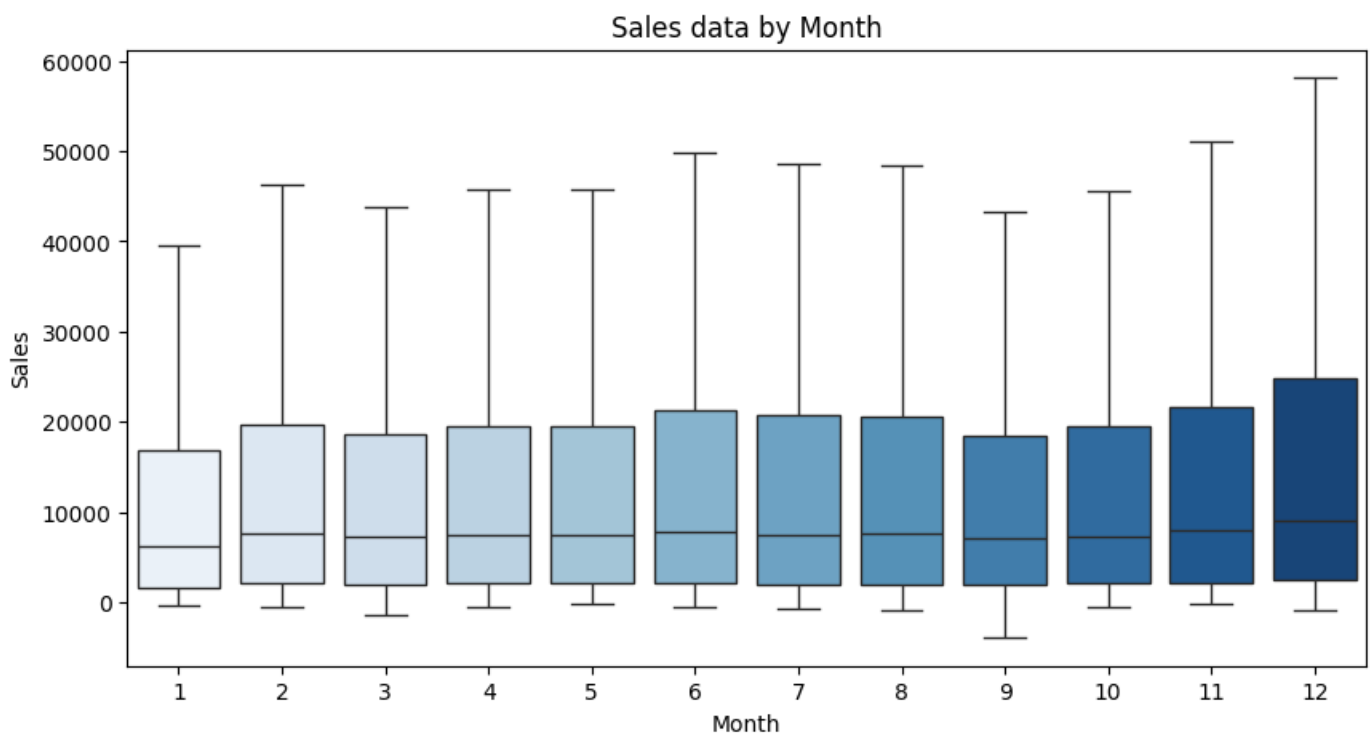
```
In [26]: pdf['Day'] = pdf.index.day
pdf['Month'] = pdf.index.month
pdf['Quarter'] = pdf.index.quarter
pdf['Year'] = pdf.index.year
```

```
In [27]: plt.figure(figsize=(10,5))
sns.boxplot(data=pdf.loc[(pdf.index > '2011-01-01') & (pdf.index < '2011-12-31')], x='Month',
            palette='Blues', showfliers=False)
            .set_title('Sales data by Month')
plt.ylabel('Sales')
plt.show()
```

<ipython-input-27-51dec4516721>:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=pdf.loc[(pdf.index > '2011-01-01') & (pdf.index < '2011-12-31')], x='Month', y='Weekly_Sales',
```



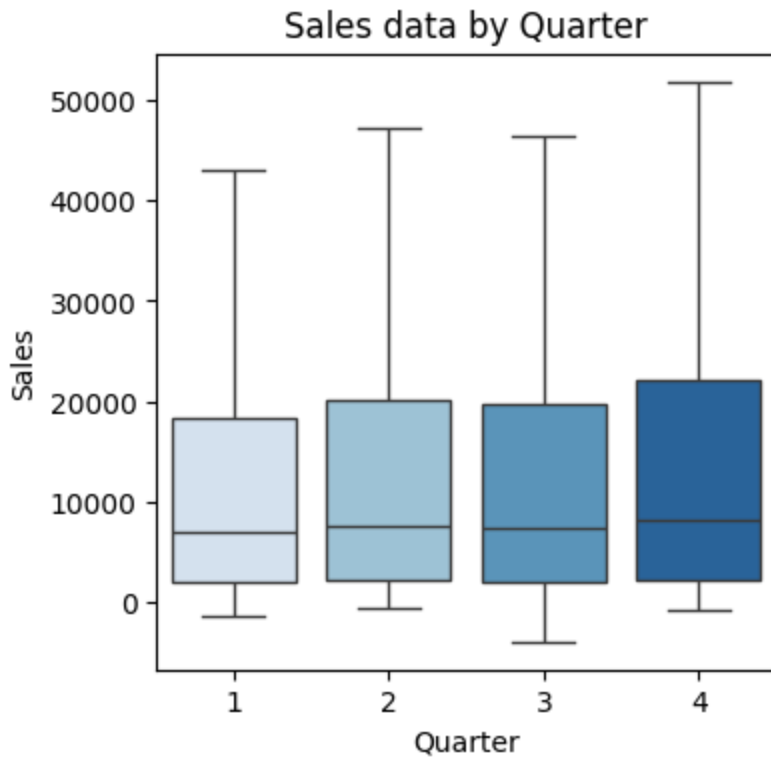
```
In [28]: plt.figure(figsize=(4,4))
```

```
sns.boxplot(data=pdf.loc[(pdf.index > '2011-01-01') & (pdf.index < '2011-12-31')], x='Quarter',
             palette='Blues', showfliers=False)
plt.set_title('Sales data by Quarter')
plt.ylabel('Sales')
plt.show()
```

<ipython-input-28-52ce5d04cfc5>:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=pdf.loc[(pdf.index > '2011-01-01') & (pdf.index < '2011-12-31')], x='Quarter', y='Weekly_Sales',
```

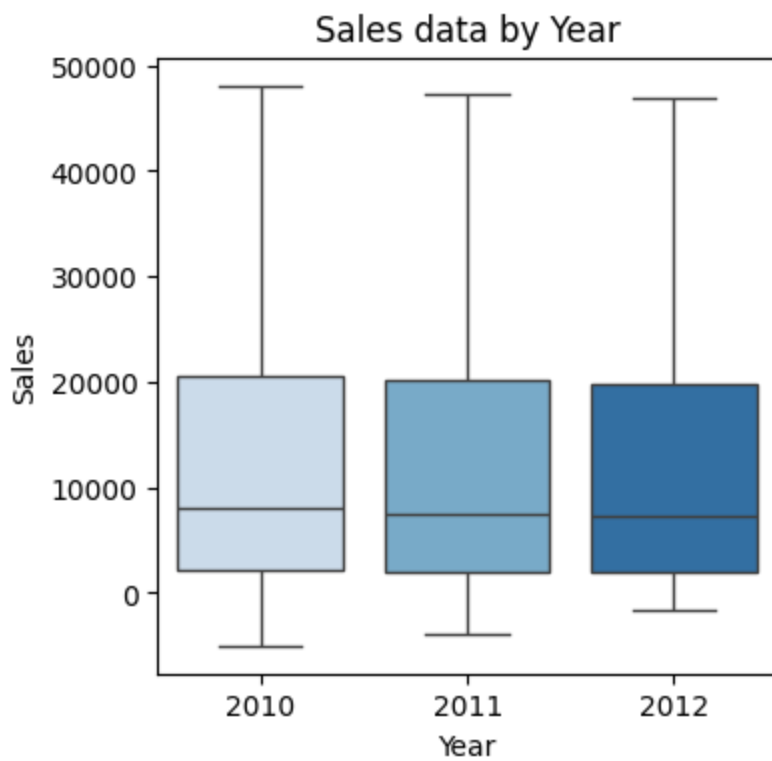


```
In [29]: plt.figure(figsize=(4,4))
sns.boxplot(data=pdf, x='Year', y='Weekly_Sales',
            palette='Blues', showfliers=False)
plt.set_title('Sales data by Year')
plt.ylabel('Sales')
plt.show()
```

<ipython-input-29-3e6635473dc7>:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=pdf, x='Year', y='Weekly_Sales',
```

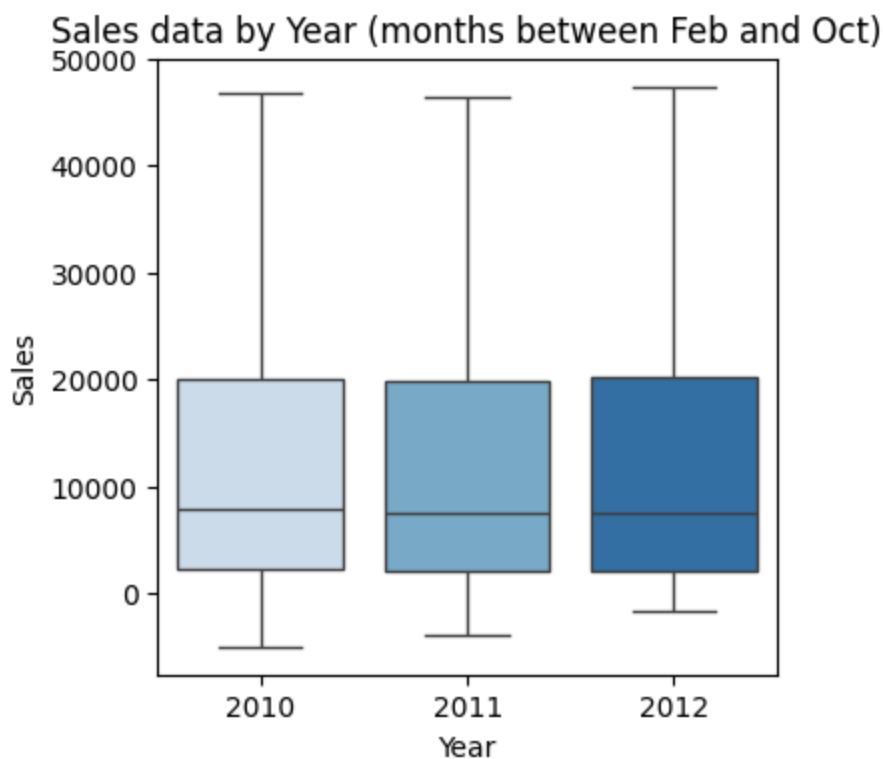


```
In [30]: plt.figure(figsize=(4,4))
sns.boxplot(data=pdf.loc[pdf.index.month.isin([2,3,4,5,6,7,8,9,10])], x='Year', y='Weekly_Sales',
            palette='Blues', showfliers=False)
plt.set_title('Sales data by Year (months between Feb and Oct)')
plt.ylabel('Sales')
plt.show()
```

<ipython-input-30-f8b0aa8dfcee>:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=pdf.loc[pdf.index.month.isin([2,3,4,5,6,7,8,9,10])], x='Year', y='Weekly_Sales',
```

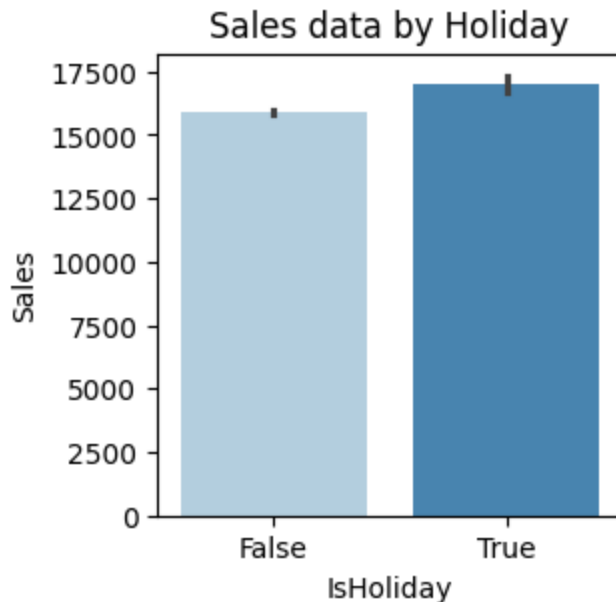


```
In [31]: plt.figure(figsize=(3,3))
sns.barplot(data=pdf, x='IsHoliday', y='Weekly_Sales',
            palette='Blues', #showfliers=False
            ).set_title('Sales data by Holiday')
plt.ylabel('Sales')
plt.show()
```

<ipython-input-31-fb602c8b9990>:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=pdf, x='IsHoliday', y='Weekly_Sales',
```

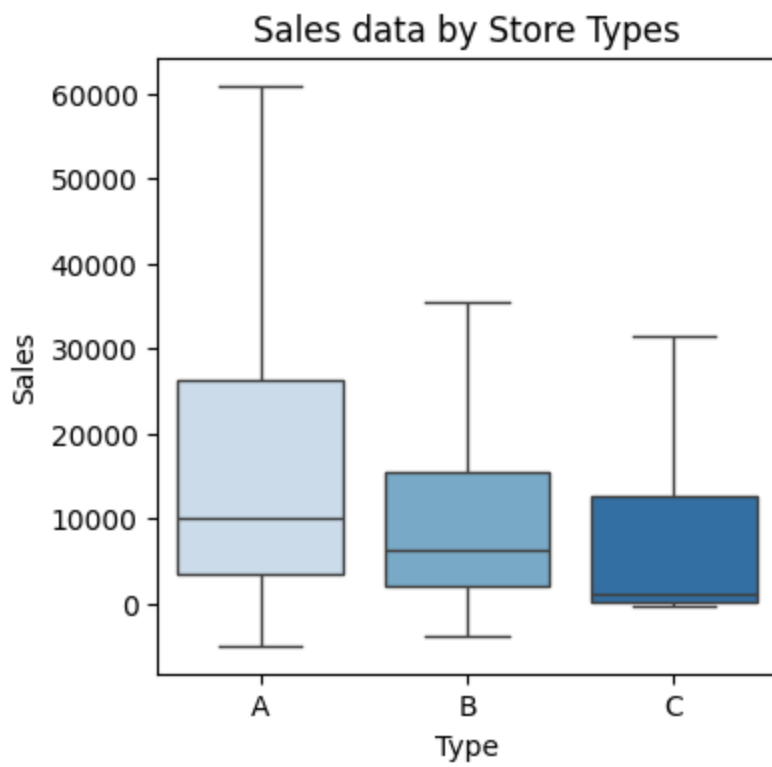


```
In [32]: plt.figure(figsize=(4,4))
sns.boxplot(data=pdf, x='Type', y='Weekly_Sales',
            palette='Blues', showfliers=False
            ).set_title('Sales data by Store Types')
plt.ylabel('Sales')
plt.show()
```

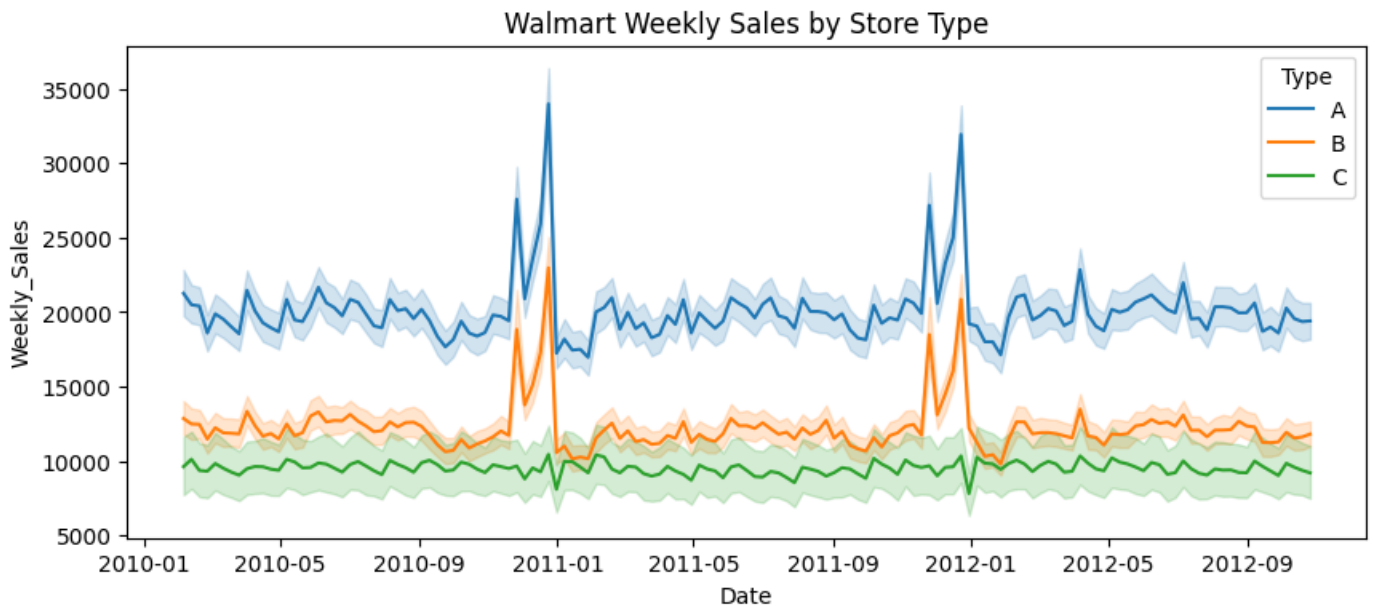
<ipython-input-32-b29283bce2fe>:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

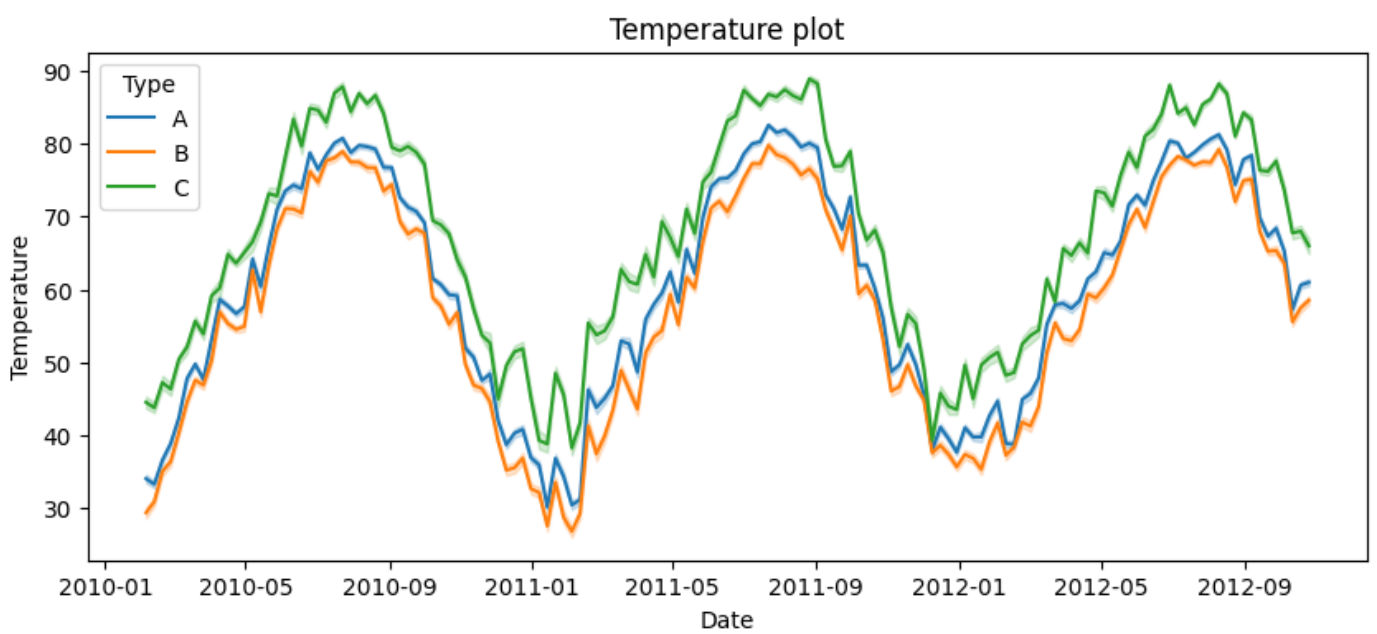
```
sns.boxplot(data=pdf, x='Type', y='Weekly_Sales',
```



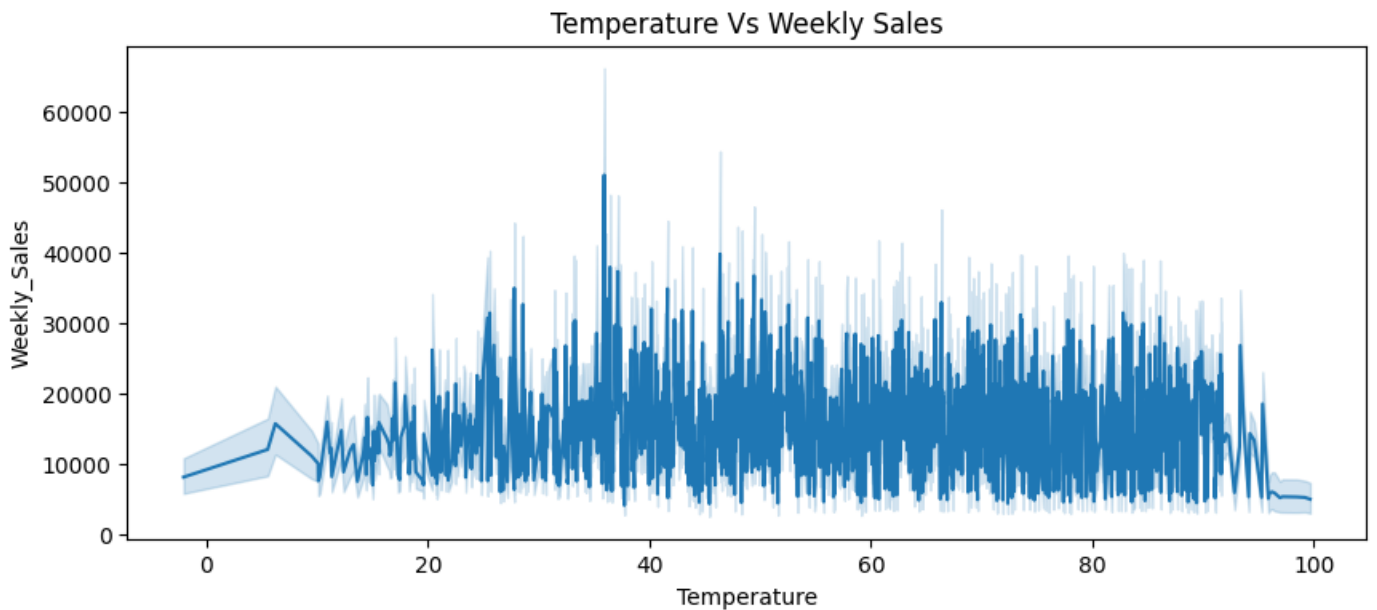
```
In [33]: plt.figure(figsize=(10,4))
sns.lineplot(data=pdf,
              x='Date',
              y='Weekly_Sales',
              hue='Type')
              ).set_title('Walmart Weekly Sales by Store Type')
plt.show()
```



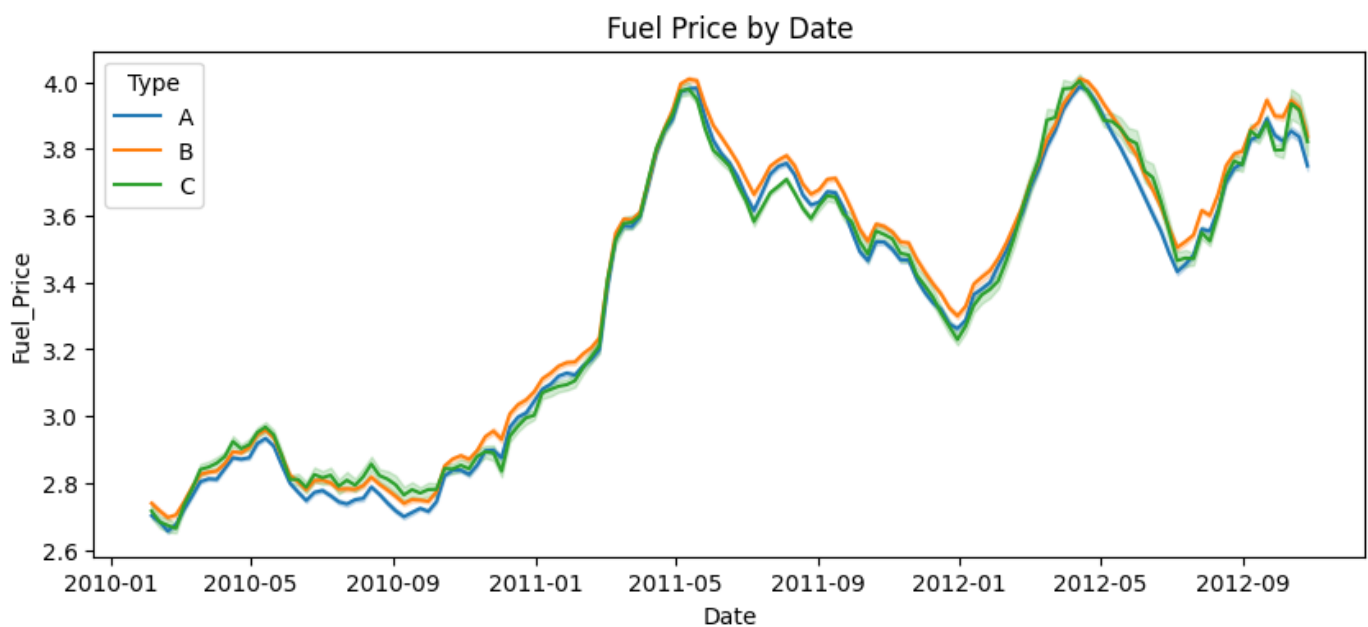
```
In [34]: fig, ax = plt.subplots(figsize=(10,4))
sns.lineplot(data=pdf,
              x='Date',
              y='Temperature',
              hue='Type',
              ax=ax,
              ).set_title('Temperature plot')
plt.show()
```



```
In [35]: plt.figure(figsize=(10,4))
sns.lineplot(data=pdf.loc[(pdf.index > '2011-01-01') & (pdf.index < '2011-12-31')],
             x='Temperature',
             y='Weekly_Sales',
             ).set_title('Temperature Vs Weekly Sales')
plt.show()
```

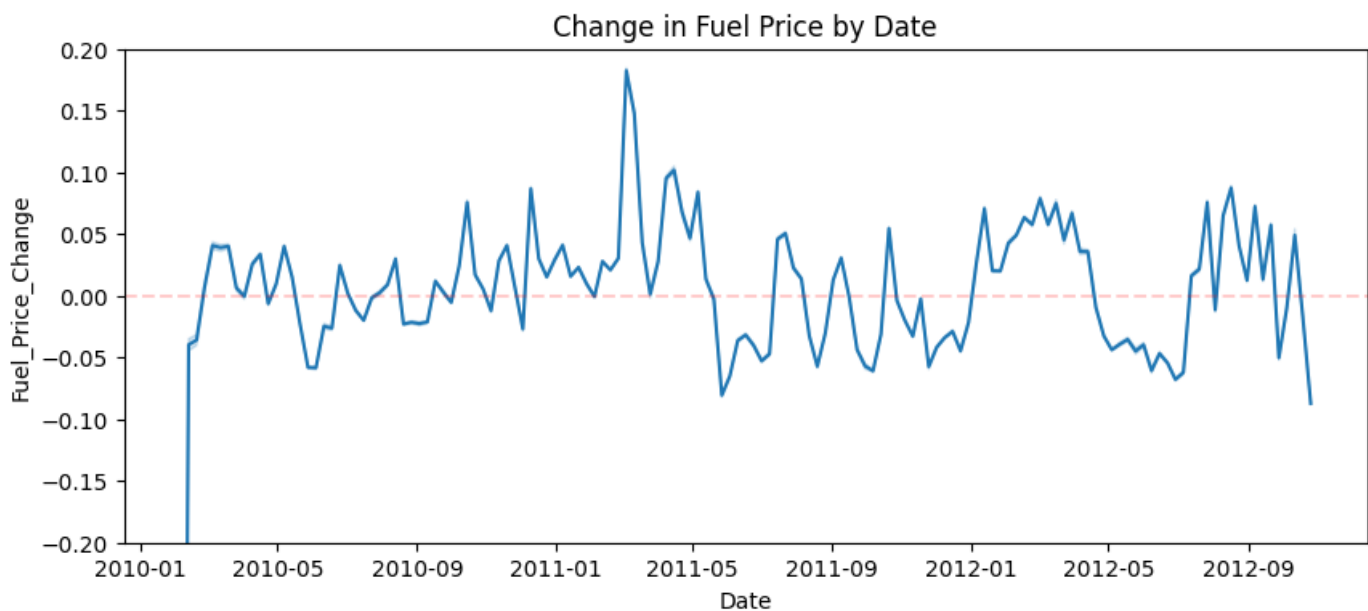


```
In [36]: plt.figure(figsize=(10,4))
sns.lineplot(data=pdf,
             x='Date',
             y='Fuel_Price',
             hue='Type',
             ).set_title('Fuel Price by Date')
plt.show()
```

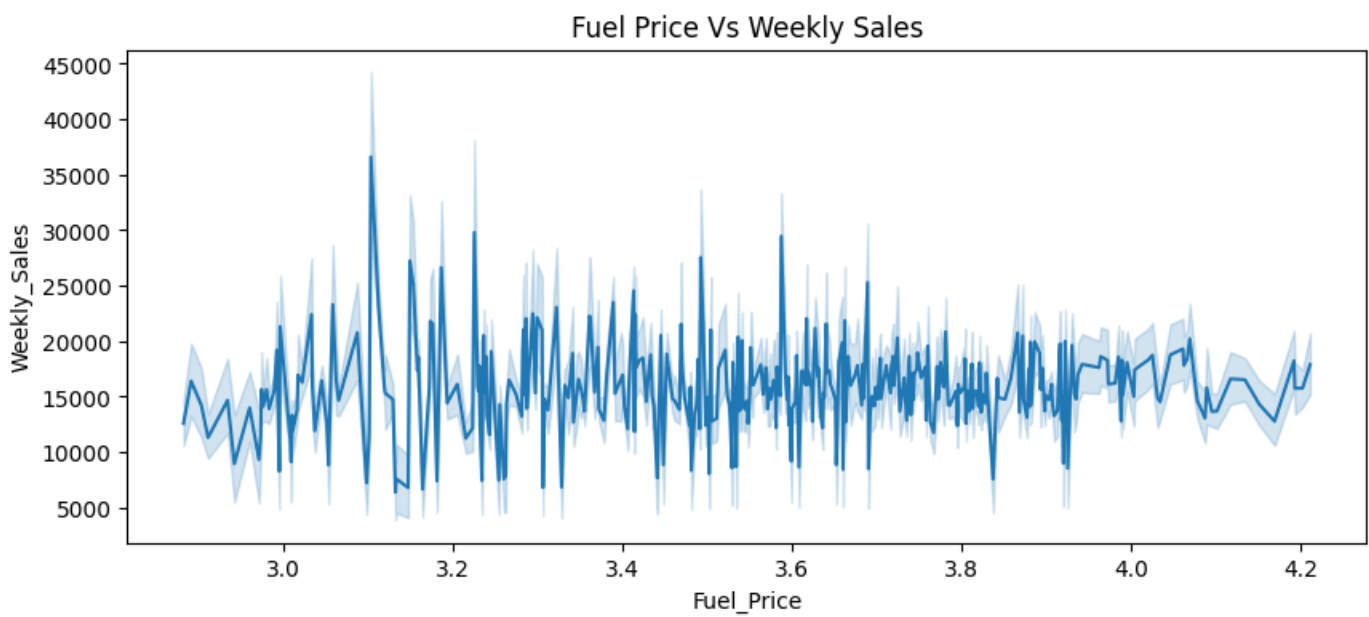



```
In [37]: pdf['Fuel_Price_Change'] = pdf['Fuel_Price'].rolling(window=2).apply(lambda x: x.iloc[1]
```

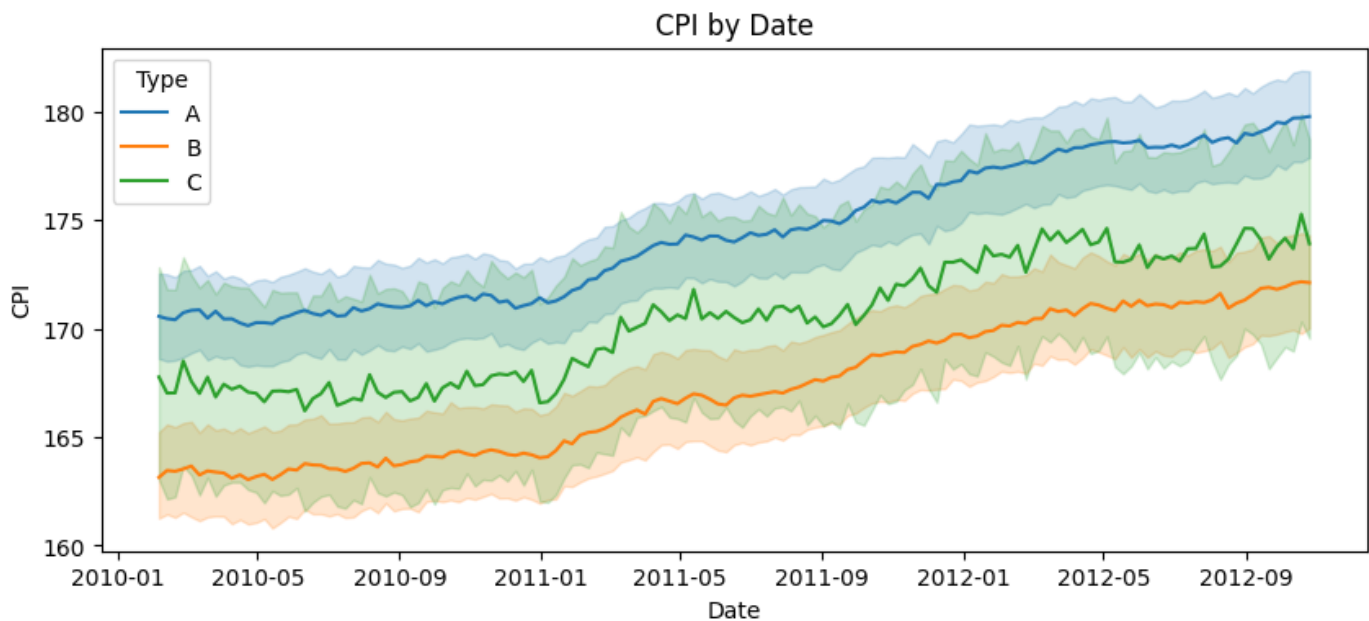
```
In [38]: plt.figure(figsize=(10,4))
sns.lineplot(data=pdf,
             x='Date',
             y='Fuel_Price_Change',
             #hue='Type',
             ).set_title('Change in Fuel Price by Date')
plt.axhline(pdf['Fuel_Price_Change'].mean(), color='r', alpha=0.2, linestyle='--')
plt.ylim(-0.2, 0.2)
plt.show()
```



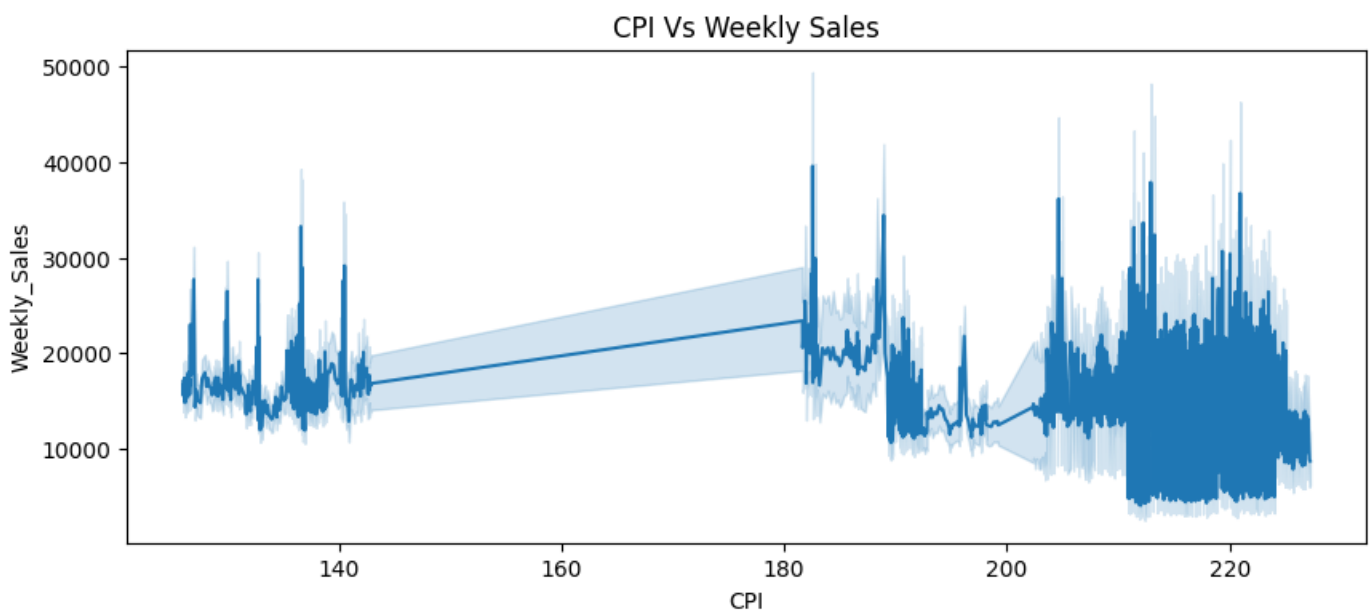
```
In [39]: plt.figure(figsize=(10,4))
sns.lineplot(data=pdf.loc[(pdf.index > '2011-01-01') & (pdf.index < '2011-12-31')],
             x='Fuel_Price',
             y='Weekly_Sales',
             ).set_title('Fuel Price Vs Weekly Sales')
plt.show()
```



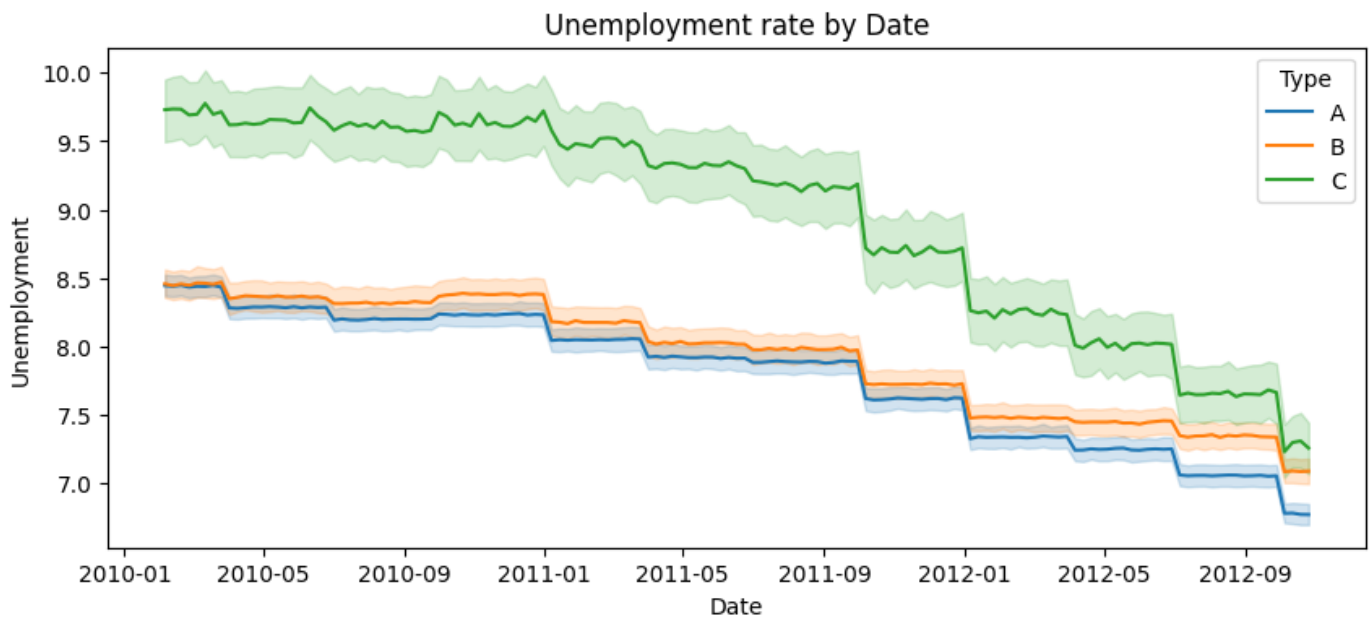
```
In [40]: plt.figure(figsize=(10,4))
sns.lineplot(data=pdf,
              x='Date',
              y='CPI',
              hue='Type')
              .set_title('CPI by Date')
plt.show()
```



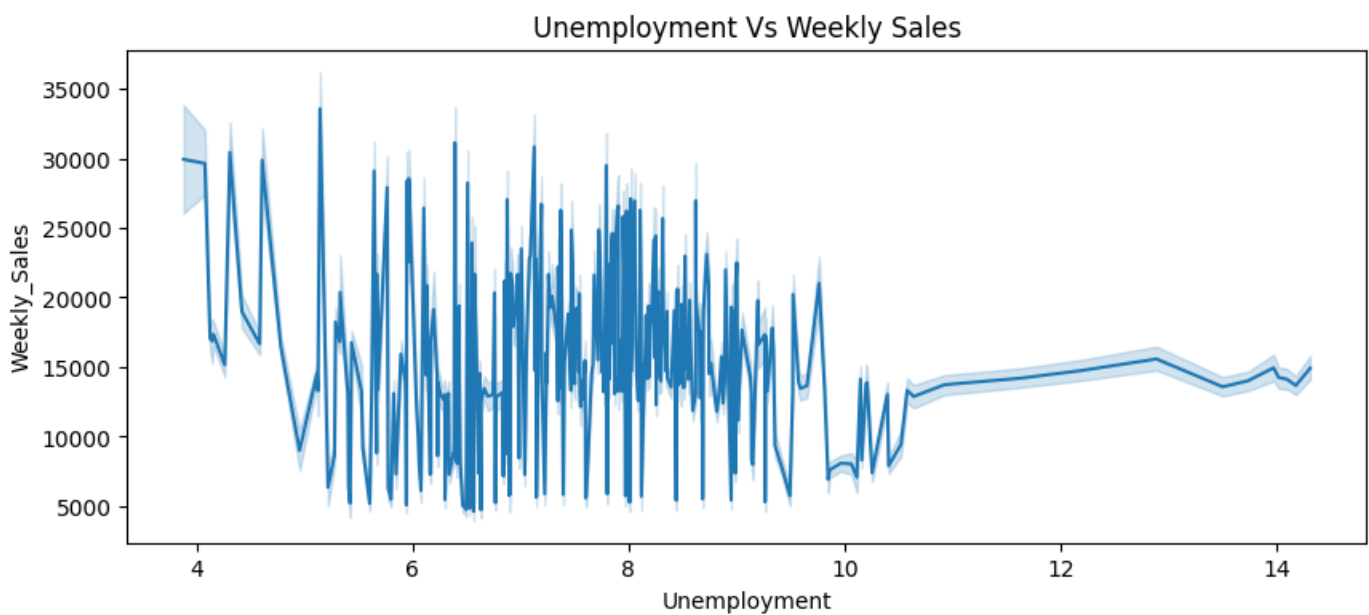
```
In [41]: plt.figure(figsize=(10,4))
sns.lineplot(data=pdf,
              x='CPI',
              y='Weekly_Sales',
              ).set_title('CPI Vs Weekly Sales')
plt.show()
```



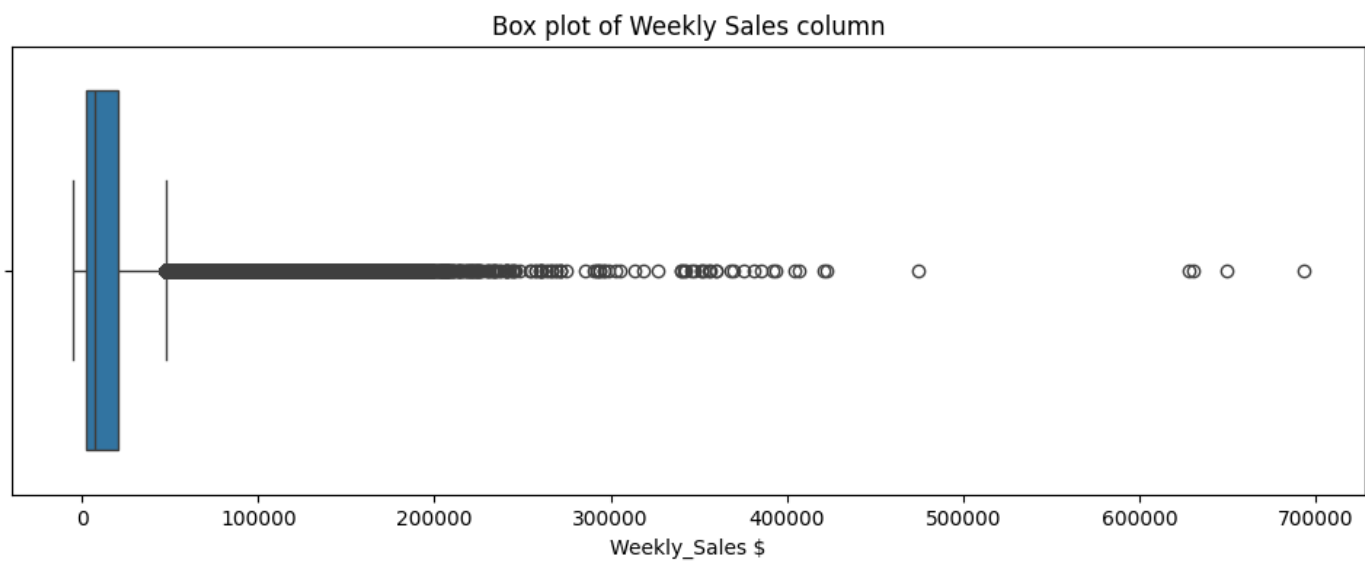
```
In [42]: plt.figure(figsize=(10,4))
sns.lineplot(data=pdf,
             x='Date',
             y='Unemployment',
             hue='Type',
             ).set_title('Unemployment rate by Date')
plt.show()
```



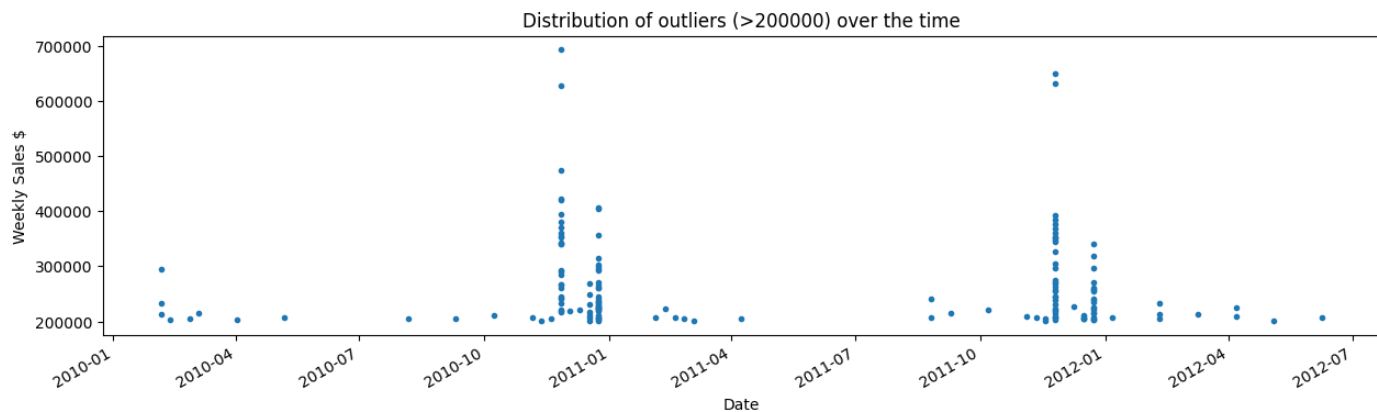
```
In [43]: plt.figure(figsize=(10,4))
sns.lineplot(data=pdf,
             x='Unemployment',
             y='Weekly_Sales',
             ).set_title('Unemployment Vs Weekly Sales')
plt.show()
```



```
In [44]: plt.figure(figsize=(12,4))
sns.boxplot(pdf['Weekly_Sales'], orient='h')
plt.xlabel('Weekly_Sales $')
plt.title('Box plot of Weekly Sales column')
plt.show()
```

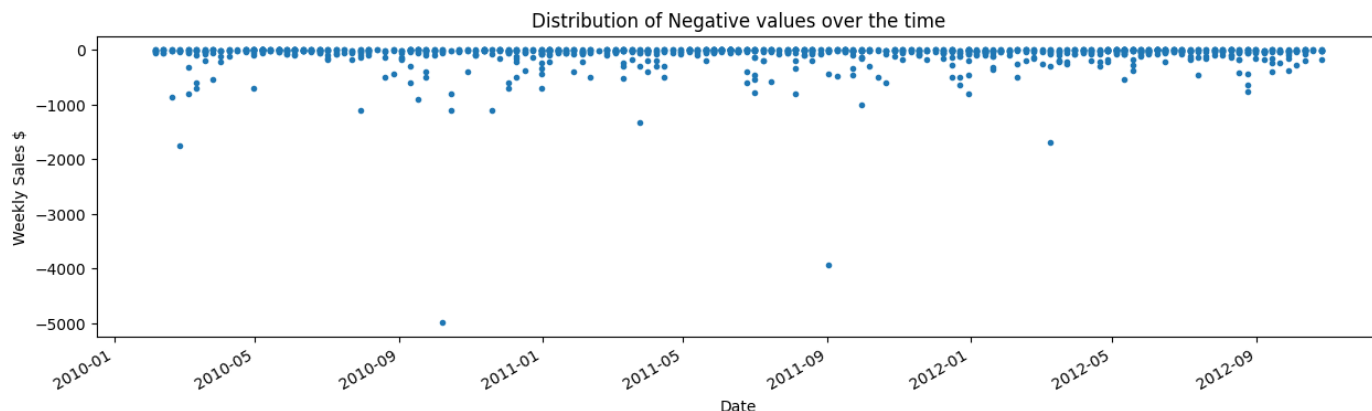


```
In [45]: pdf.query('Weekly_Sales > 200000')['Weekly_Sales'].plot(figsize=(15,4), style='.')
plt.ylabel('Weekly Sales $')
plt.title('Distribution of outliers (>200000) over the time')
plt.show()
```



```
In [46]: pdf.query('Weekly_Sales < 0')['Weekly_Sales'].plot(figsize=(15,4), style='.')
```

```
plt.ylabel('Weekly Sales $')
plt.title('Distribution of Negative values over the time')
plt.show()
```



```
In [47]: old_count = pdf.shape[0]
print('Dataset_size before removing outliers: ', old_count)
pdf = pdf.query('Weekly_Sales >= 0 and Weekly_Sales < 200000')
new_count = pdf.shape[0]
print('Dataset_size after removing outliers: ', pdf.shape[0])
print('Percent decrease in data: ', round((old_count - new_count)*100/old_count,2))
```

```
Dataset_size before removing outliers: 421570
Dataset_size after removing outliers: 420122
Percent decrease in data: 0.34
```

```
In [48]: # Numerical cols
num_cols = [x for x in pdf.columns if x not in ['Type', 'IsHoliday', 'Dept', 'Store']]

# Compute the correlation matrix
corr = pdf[num_cols].corr().round(3)

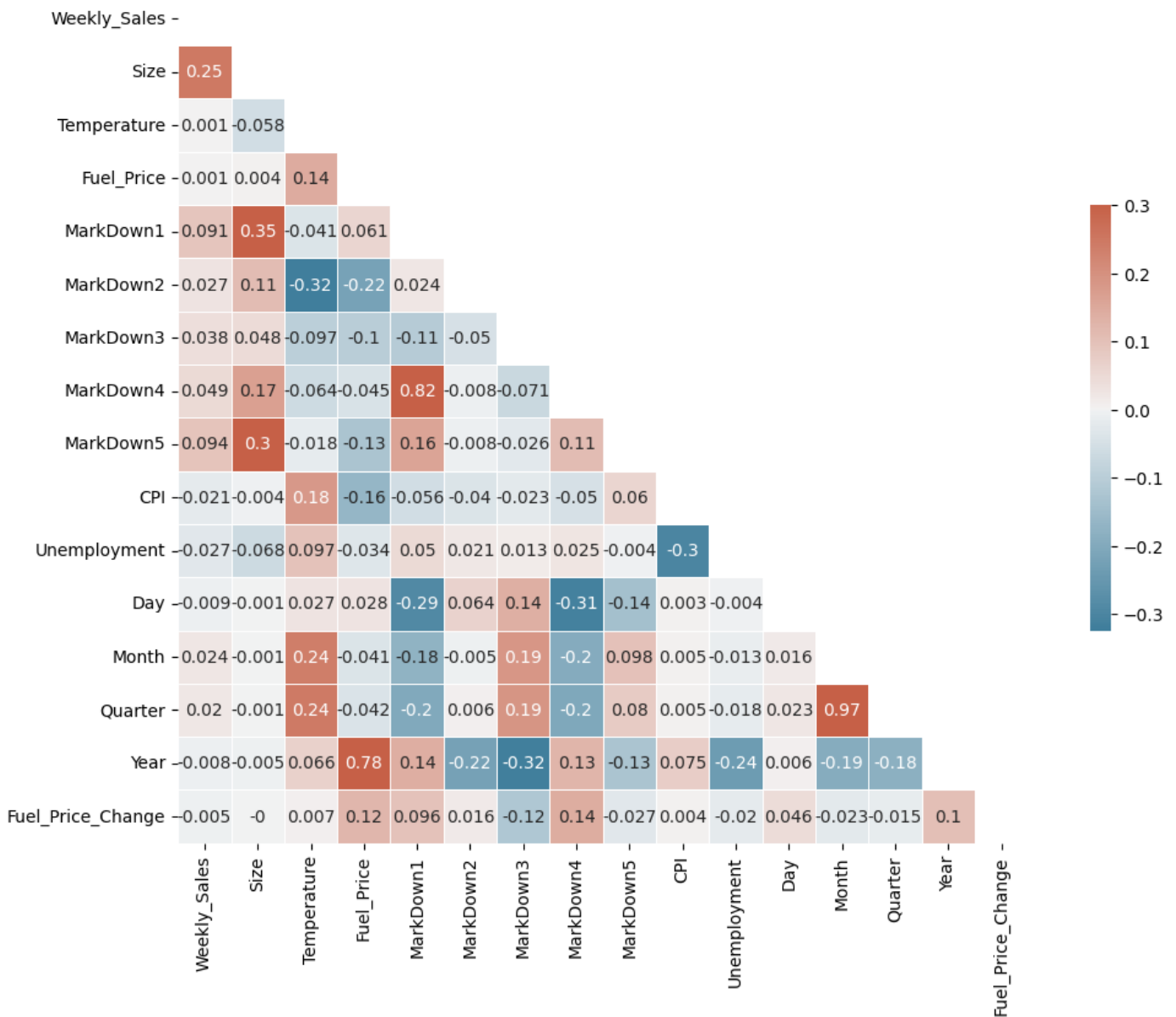
# Generate a mask for the upper triangle
mask = np.triu(np.ones_like(corr, dtype=bool))

# Set up the matplotlib figure
f, ax = plt.subplots(figsize=(13, 9))

# Generate a custom diverging colormap
cmap = sns.diverging_palette(230, 20, as_cmap=True)

# Draw the heatmap with the mask and correct aspect ratio
sns.heatmap(corr, mask=mask, cmap=cmap, vmax=.3, center=0,
            square=True, annot=True, linewidths=.5, cbar_kws={"shrink": .5})
plt.title('Correlation Matrix')
plt.show()
```

Correlation Matrix



```
In [49]: #Preprocssing data
!pip install xgboost
```

Requirement already satisfied: xgboost in /usr/local/lib/python3.10/dist-packages (2.0.3)
Requirement already satisfied: numpy in /usr/local/lib/python3.10/dist-packages (from xgboost) (1.25.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.10/dist-packages (from xgboost) (1.11.4)

```
In [52]: #Joining the dataset and creating features
```

```
from pyspark.sql.functions import when, col

df = spark.read.options(delimiter=',', inferSchema=True, header=True).csv('/content/driv
stores = spark.read.options(delimiter=',', inferSchema=True, header=True).csv('/content/
features = spark.read.options(delimiter=',', inferSchema=True, header=True).csv('/conten
# Converting 'NA' strings in MarkDown columns to null values
for i in range(1, 6):
    features = features.withColumn(f'MarkDown{i}', when(col(f'MarkDown{i}')=='NA', None))

# Joining DataFrames
pdf = df.join(stores, on='Store', how='inner').join(features.drop('IsHoliday'), on=['Dat

# Further processing with pandas
```

```
pdf['Date'] = pd.to_datetime(pdf['Date'])
pdf.set_index('Date', inplace=True)
pdf["CPI"] = pd.to_numeric(pdf["CPI"], downcast="float")
pdf["Unemployment"] = pd.to_numeric(pdf["Unemployment"], downcast="float")
pdf['Day'] = pdf.index.day
pdf['Week'] = np.ceil(pdf.index.dayofyear / 7).astype(int)
pdf['Month'] = pdf.index.month
pdf['Quarter'] = pdf.index.quarter
pdf['Year'] = pdf.index.year
pdf = pdf.query('Weekly_Sales >= 0 and Weekly_Sales < 200000')

# Display the first few rows
pdf.head().style.hide(axis='index')
```

```
Out[52]:
```

Store	Dept	Weekly_Sales	IsHoliday	Type	Size	Temperature	Fuel_Price	MarkDown1	MarkDown2	MarkD
1	1	24924.500000	False	A	151315	42.310000	2.572000	nan	nan	
1	1	46039.490000	True	A	151315	38.510000	2.548000	nan	nan	
1	1	41595.550000	False	A	151315	39.930000	2.514000	nan	nan	
1	1	19403.540000	False	A	151315	46.630000	2.561000	nan	nan	
1	1	21827.900000	False	A	151315	46.500000	2.625000	nan	nan	

```
In [53]: pdf.isnull().sum()
```

```
Out[53]:
```

Store	0
Dept	0
Weekly_Sales	0
IsHoliday	0
Type	0
Size	0
Temperature	0
Fuel_Price	0
MarkDown1	269992
MarkDown2	309270
MarkDown3	283525
MarkDown4	285656
MarkDown5	269244
CPI	0
Unemployment	0
Day	0
Week	0
Month	0
Quarter	0
Year	0
dtype:	int64

```
In [54]: pdf.fillna(0, inplace=True)
```

```
In [55]: #Converting String / Boolean Columns into Numerical
#Converting "Is Holiday" column to int
#Changing the type will replace True -> 1 and False -> 0

pdf['IsHoliday'] = pdf['IsHoliday'].astype('int')
```

```
In [56]: # Converting Type of store into numerical
# We will use the order of size to decide the numerical values for store types
# As we have seen Type A stores are largest and C are smallest,
# we will use the following mapping
# C-> 0, B-> 1, A-> 2

pdf['Type'] = pdf['Type'].replace({'C': 0, 'B': 1, 'A': 2})
```

```
In [57]: #XG Boost

TARGET = ['Weekly_Sales']
FEATURES = [i for i in pdf.columns if i not in TARGET]
```

```
In [58]: from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(pdf[FEATURES], pdf[TARGET],
                                                    test_size=0.2, random_state=130)
```

```
In [59]: import xgboost as xgb
reg = xgb.XGBRegressor(n_estimators=2000, early_stopping_rounds=50,
                       learning_rate=0.5)
model = reg.fit(X_train, y_train,
                eval_set=[(X_train, y_train), (X_test, y_test)],
                verbose=100)

[0]      validation_0-rmse:15063.79170      validation_1-rmse:15139.93328
[100]     validation_0-rmse:4242.24583      validation_1-rmse:4492.70623
[200]     validation_0-rmse:3482.17459      validation_1-rmse:3878.64994
[300]     validation_0-rmse:3087.08542      validation_1-rmse:3587.85615
[400]     validation_0-rmse:2811.11912      validation_1-rmse:3408.92916
[500]     validation_0-rmse:2608.76790      validation_1-rmse:3287.69013
[600]     validation_0-rmse:2453.39392      validation_1-rmse:3205.62576
[700]     validation_0-rmse:2334.43202      validation_1-rmse:3135.30009
[800]     validation_0-rmse:2237.46742      validation_1-rmse:3089.50207
[900]     validation_0-rmse:2146.26464      validation_1-rmse:3049.58338
[1000]    validation_0-rmse:2069.41808      validation_1-rmse:3019.87896
[1100]    validation_0-rmse:2000.64612      validation_1-rmse:2989.92329
[1200]    validation_0-rmse:1937.12835      validation_1-rmse:2962.81252
[1300]    validation_0-rmse:1882.34812      validation_1-rmse:2942.05732
[1400]    validation_0-rmse:1837.24162      validation_1-rmse:2926.77431
[1500]    validation_0-rmse:1783.89858      validation_1-rmse:2904.19247
[1600]    validation_0-rmse:1741.64163      validation_1-rmse:2893.16655
[1700]    validation_0-rmse:1701.23306      validation_1-rmse:2880.83820
[1800]    validation_0-rmse:1663.19122      validation_1-rmse:2870.64874
[1900]    validation_0-rmse:1628.06565      validation_1-rmse:2860.12257
[1999]    validation_0-rmse:1594.79638      validation_1-rmse:2852.11122
```

```
In [60]: y_pred = reg.predict(X_test)
```

```
In [62]: from sklearn import metrics
import numpy as np

xgboost_accuracy = reg.score(X_test, y_test)*100
print("XGB Regressor Evaluation Metrics: ")
print("Accuracy", round(xgboost_accuracy,3))
print("MAE \t" , round(metrics.mean_absolute_error(y_test, y_pred),3))
print("MSE \t" , round(metrics.mean_squared_error(y_test, y_pred),3))
print("RMSE \t" , round(np.sqrt(metrics.mean_squared_error(y_test, y_pred)),3))
print("R2 \t" , round(metrics.explained_variance_score(y_test, y_pred),5))
```

```
XGB Regressor Evaluation Metrics:
Accuracy 98.348
MAE      1570.744
MSE      8134464.931
RMSE     2852.098
R2       0.98348
```

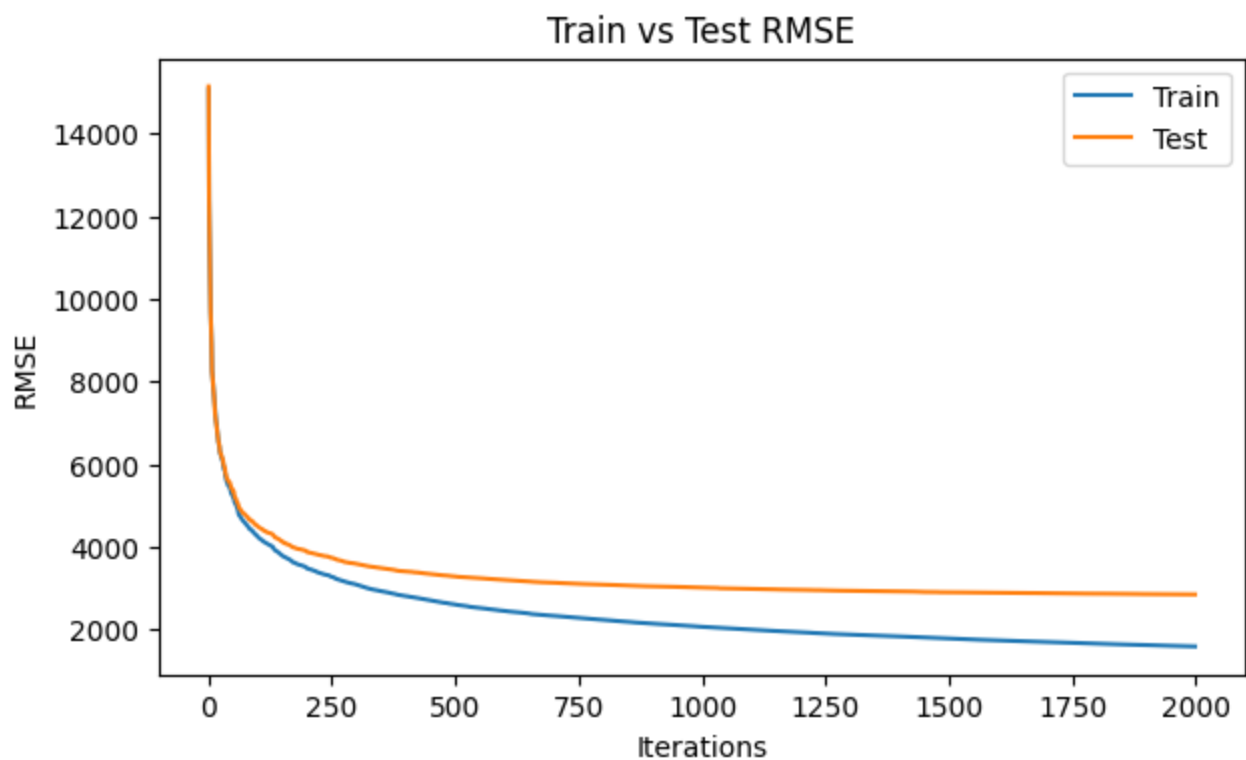
```
In [63]: print('Top 10 features with their Feature importance scores: ')
pd.DataFrame(data=reg.feature_importances_,
             index=reg.feature_names_in_,
             columns=['Feature Importance']).sort_values('Feature Importance', ascending=
```

Top 10 features with their Feature importance scores:

```
Out[63]:      Feature Importance
```


Size	0.349576
Dept	0.214630
Type	0.146933
Store	0.098718
CPI	0.038189
Week	0.030618
IsHoliday	0.024891
Month	0.023361
Year	0.019829
Unemployment	0.011220

```
In [64]: fig, ax = plt.subplots(1,1,figsize=(7,4))
score_hist = pd.DataFrame({'Train_rmse':model.evals_result()['validation_0']['rmse'], 'Test_rmse':model.evals_result()['validation_1']['rmse']})
sns.lineplot(score_hist['Train_rmse'], ax=ax, label='Train')
sns.lineplot(score_hist['Test_rmse'], ax=ax, label='Test')
plt.ylabel('RMSE')
plt.xlabel('Iterations')
plt.title('Train vs Test RMSE')
plt.show()
```

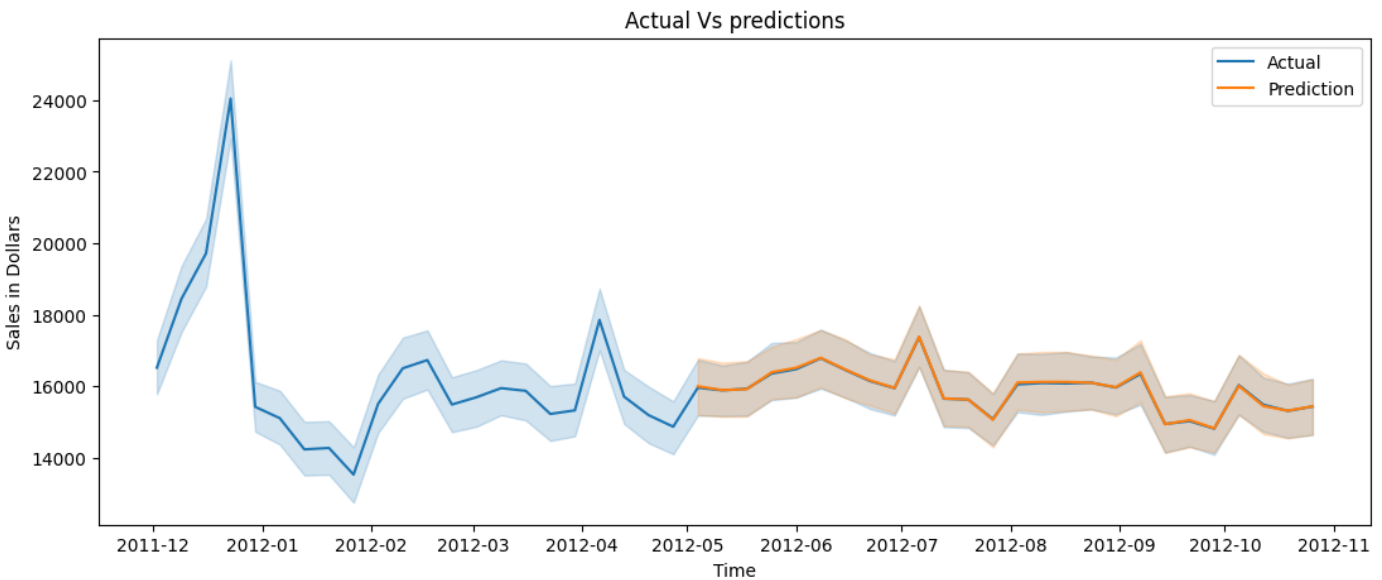


```
In [65]: val = pdf.loc[pdf.index >= '2012-05-01']
X_val, y_val = val[FEATURES], val[TARGET]
val['prediction'] = reg.predict(X_val)
```

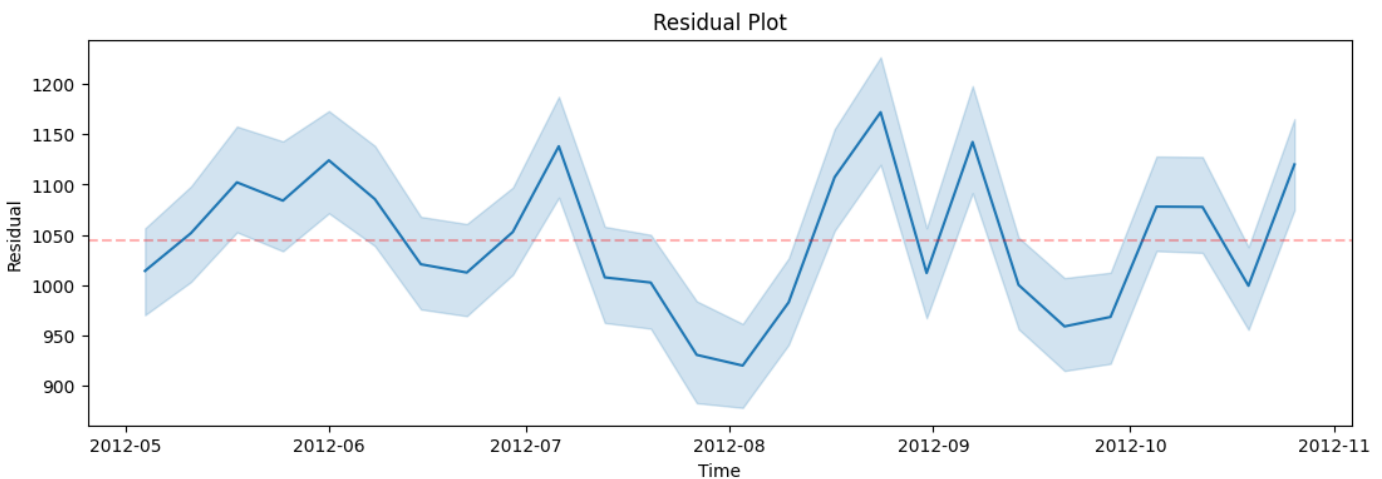
<ipython-input-65-315f3b7bbeb8>:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
val['prediction'] = reg.predict(X_val)

```
In [66]: fig, ax = plt.subplots(1,1,figsize=(13,5))
sns.lineplot(pdf['Weekly_Sales'].loc[pdf.index>'2011-12-01'], ax=ax, label='Actual')
sns.lineplot(val['prediction'], ax=ax, label='Prediction')
plt.ylabel('Sales in Dollars')
plt.xlabel('Time')
plt.title('Actual Vs predictions')
plt.show()
```



```
In [67]: val = val.assign(Residual= abs(val['Weekly_Sales'] - val['prediction']))
fig, ax = plt.subplots(1,1,figsize=(13,4))
#sns.lineplot(pdf['Weekly_Sales'].loc[pdf.index>'2011-12-01'], ax=ax, label='Actual')
sns.lineplot(val['Residual'], ax=ax)
plt.axhline(val['Residual'].mean(), linestyle='--', color='r', alpha=0.3)
#plt.ylabel('Sales in Dollars')
plt.xlabel('Time')
plt.title('Residual Plot')
plt.show()
```



```
In [68]: # Linear Regression
from sklearn.linear_model import LinearRegression
linReg = LinearRegression()
linReg.fit(X_train,y_train)

linReg_predicted = linReg.predict(X_test)
```

```
In [72]: linear_accuracy = linReg.score(X_test,y_test)*100
print("Linear Regression Evaluation Metrics: ")
# print("Accuracy", round(linear_accuracy,3))
print("Accuracy" , linear_accuracy)
```

```
print("MAE \t", round(metrics.mean_absolute_error(y_test, linReg_predicted),3))
print("MSE \t", round(metrics.mean_squared_error(y_test, linReg_predicted),3))
print("RMSE \t", round(np.sqrt(metrics.mean_squared_error(y_test, linReg_predicted)),3))
print("R2 \t", round(metrics.explained_variance_score(y_test, linReg_predicted),5))
```

Linear Regression Evaluation Metrics:

```
Accuracy 8.828598202030758
MAE      14485.662
MSE      448938273.001
RMSE     21188.164
R2       0.08829
```

```
In [74]: from sklearn.ensemble import RandomForestRegressor
randomForest = RandomForestRegressor(n_estimators=30)
randomForest.fit(X_train, y_train)
randomForest_predict = randomForest.predict(X_test)

rf_accuracy = randomForest.score(X_test,y_test)*100
print("Random Forest Evaluation Metrics: ")
print("Accuracy", round(rf_accuracy,3))
print("MAE \t", round(metrics.mean_absolute_error(y_test, randomForest_predict),3))
print("MSE \t", round(metrics.mean_squared_error(y_test, randomForest_predict),3))
print("RMSE \t", round(np.sqrt(metrics.mean_squared_error(y_test, randomForest_predict)),3))
print("R2 \t", round(metrics.explained_variance_score(y_test, randomForest_predict),5))
```

<ipython-input-74-3af473820916>:3: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

```
randomForest.fit(X_train, y_train)
```

Random Forest Evaluation Metrics:

```
Accuracy 98.021
MAE      1336.029
MSE      9745770.756
RMSE     3121.822
R2       0.98021
```

```
In [84]: # Performing GridSearchCV on Decision Tree Regression
from sklearn.tree import DecisionTreeRegressor
from sklearn.model_selection import GridSearchCV

depth = list(range(3,11))
param_grid = dict(max_depth = depth)
tree = GridSearchCV(DecisionTreeRegressor(), param_grid, cv = 10)
tree.fit(X_train,y_train)
```

```
Out[84]: □ GridSearchCV
□ estimator: DecisionTreeRegressor
□ DecisionTreeRegressor
```

```
In [85]: # Predicting train and test results
from math import sqrt
from sklearn.metrics import r2_score
from sklearn.metrics import mean_squared_error as mse

y_train_pred = tree.predict(X_train)
y_test_pred = tree.predict(X_test)

dt_accuracy = tree.score(X_test,y_test)*100
print("Accuracy", dt_accuracy)
print("Train Results for Decision Tree Regressor Model:")
print("Root Mean squared Error: ", sqrt(mse(y_train.values, y_train_pred)))
print("R-Squared: ", r2_score(y_train.values, y_train_pred))
```

```
print("Test Results for Decision Tree Regressor Model:")
print("Root Mean Squared Error: ", sqrt(mse(y_test, y_test_pred)))
print("R-Squared: ", r2_score(y_test, y_test_pred))

# print("Decision Tree Evaluation Metrics: ")
# print("Accuracy", round(dt_accuracy,3))
# print("MAE \t" , round(metrics.mean_absolute_error(y_test, randomForest_predict),3))
# print("MSE \t" , round(metrics.mean_squared_error(y_test, randomForest_predict),3))
# print("RMSE \t" , round(np.sqrt(metrics.mean_squared_error(y_test, randomForest_predict),3))
# print("R2 \t" , round(metrics.explained_variance_score(y_test, randomForest_predict),5))
```

Accuracy 87.81936258491231

Train Results for Decision Tree Regressor Model:

Root Mean squared Error: 7662.26261676269

R-Squared: 0.8800314587484894

Test Results for Decision Tree Regressor Model:

Root Mean Squared Error: 7744.6002779157425

R-Squared: 0.8781936258491231